Kenneth W. Bingham

Senior Full-Stack Developer | AI-Directed Engineer

St. George, UT (Mountain Time) | Remote, or hybrid in St. George | Contact: butterflyfx.us/contact.html
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Portfolio: butterflyfx.us | Games: playbutterflyfx.com | Gallery: kensartgallery.com | Videos: youtube.com/@kbingh | Substack: substack.com/@kbingh

SUMMARY

Senior full-stack developer and AI-directed engineer with 15+ years building enterprise, government, financial and logistics systems in Java, Spring Boot, C#, .NET Core, Angular, SQL and Azure. Builds adversarial AI benchmarks and evaluation harnesses, and directs AI as an engineering lead with tests that try to break each claim.

SKILLS

Languages and frameworks: Java, Spring Boot, C#, .NET Core, Python, JavaScript, TypeScript, Angular, AngularJS, React, Node.js, FastAPI

Java and design: JPA and Hibernate (entities, @ManyToMany, lazy and eager loading), Spring Data repositories, REST controllers, MVC; design patterns: Singleton, Builder, Factory, Observer, publish-subscribe

AI engineering: Claude Code, agentic workflows, custom MCP server, RAG with pgvector, embeddings, prompt engineering, LLM evaluation (pass@k), hallucination mitigation, Ollama

Databases: SQL Server, Oracle, PostgreSQL, MySQL, MongoDB, MarkLogic; query optimization, schema design

Cloud and DevOps: Azure, Azure DevOps, AWS, Cloudflare Workers, CI/CD, Docker, Ubuntu Server (install and full management), Apache, nginx, Tomcat, Git, TFS, Maven, NuGet

3D and web: Three.js, WebGL, WebXR, responsive design, technical SEO; voxel grids, octrees, spatial hashing

Testing and debugging: JUnit, NUnit, xUnit, MSTest, Selenium, Postman, Swagger/OpenAPI, Puppeteer, browser dev tools, IDE debuggers, Splunk

EXPERIENCE

AI Engineer, Evaluation and Reliability (Contract)

Handshake AI, Remote | Jul 2026 to Present

- Author adversarial benchmark tasks for Terminal-Bench 2 that defeat frontier coding agents, verified at pass@5 0/5: the reference agent at maximum reasoning effort fails every attempt.
- Built an offline difficulty-certification harness (candidate generator, CP-SAT constraint-solver filter and a blind cold-agent pass@2 run) that certifies task difficulty in minutes instead of a 68-minute CI cycle.
- Wrote a deterministic pre-submit linter encoding 104 authoring rules that catches solution leaks, ambiguous framing and self-verifiable traps before a task reaches review.

Software Developer

DTSI, Hill Air Force Base, UT | Apr 2024 to May 2026

- Principal developer on Air Force logistics and operational systems in Java, Spring Boot, Hibernate, Angular and Bootstrap, and in C#, .NET Core Razor Pages and SQL Server, under strict DoD change control.
- Configured CI/CD pipelines and secure deployment workflows with TFS version control, with zero rollbacks.

Software Developer

Humana, Remote | Feb 2022 to Jan 2024

- Built .NET 6 services in C# and Visual Studio, and HL7 and FHIR interoperability pipelines for high-volume insurance claims under HIPAA.
- Modernized legacy applications from .NET Framework to .NET Core (.NET 6) for performance and cross-platform support.

Software Developer

Software Implementers, Hill Air Force Base, UT | Apr 2021 to Dec 2021

- Developed Java, Spring, Spring Boot and Hibernate services and responsive Angular and Bootstrap front ends for aircraft-parts testing, maintenance and asset tracking.
- Turned the testers' manual spreadsheets into a central web-based CMS and workflow platform for parts tracking, test protocols and maintenance.

Software Developer (Contract via ConsultNet)

Client: The Church of Jesus Christ of Latter-day Saints, Salt Lake City, UT | Sep 2020 to Oct 2020

- Contributed to an internal application for tracking and versioning Church leaders' talks, in Java, Spring Boot, Hibernate, JSP, Thymeleaf and Bootstrap.

Software Developer

Gwinnett County Public Schools, Lawrenceville, GA | Feb 2020 to Aug 2020

- Built automated test frameworks in Java and Selenium WebDriver for the district's cloud platforms, with Azure DevOps pipelines.

Contract and agency roles

Norfolk Southern and IBM (via Collabera), InComm, Finicity (Mastercard, via STG), EFS (a WEX company) | 2017 to 2019

- Spring Boot and TIBCO messaging for locomotive fuel efficiency; Selenium WebDriver automation with grid parallel testing; payment and prepaid card systems; financial data aggregation microservices; a fleet gas-card portal.

Java Developer

CUProdigy, Layton, UT | Apr 2015 to Feb 2017

- Developed the credit union core system for small and mid-size credit unions: teller applications, management and operations, record keeping, account management, loan processing, interest calculation, and payment and collection management.

Software Consultant

Software Technology Group, Salt Lake City, UT | Sep 2011 to Jun 2014

- Missionary Portal for The Church of Jesus Christ of Latter-day Saints, used by thousands of missionaries worldwide: helped plan the architecture and built most of the UI (Bootstrap, Backbone, Handlebars) with Java Spring services and MarkLogic.
- A translation application for the same client (Java Swing, Flex), and CourtView, a case and docket application used by courts nationwide (Java, Spring, DataTables).

Contract roles

Axis41 via ConsultNet (Adobe AEM for O.C. Tanner and Stanford Health Care, Nov 2014 to Apr 2015); USANA Health Sciences via TEKsystems (AngularJS, Jun to Oct 2014) | Jun 2014 to Apr 2015

Earlier experience

Boresight (internship, Jun 2010 to Sep 2011: Java Swing avionics); Juniper Networks (college role, 2008 to 2009: networking support) | 2008 to Sep 2011

SELECTED PROJECTS (HUMAN AND AI COLLABORATION)

In each project I set the vision, design, standards and priorities and verified every result; an AI coding assistant wrote and tested the code under my direction.

- **Chomp 3D: Wally's Cairns Adventure, playbutterflyfx.com/chomp.** Seven-level 3D browser game for desktop, phone and VR (WebXR). 1,022 unit checks and 107 browser tests gate 14 versioned releases with one-command Cloudflare rollback; Workers leaderboard and feedback form.
- **citeline, github.com/kenbin64/citeline.** Retrieval-augmented generation over federal regulations with PostgreSQL, pgvector embeddings and FastAPI: answers with a citation or abstains. 31 unit tests and a 28-case evaluation set; a live version runs on Cloudflare Workers.

EDUCATION

B.S., Computer Science and Software Engineering, Weber State University, Ogden, UT, 2011

SECURITY CLEARANCE

U.S. Air Force Secret clearance (previously held), with years of delivery inside DoD environments at Hill Air Force Base.

INTERESTS

3D printing, game development, AI art, drumming, painting, singing.